

Module Three:

Proof Construction & Logical Reasoning

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CS 505: Discrete Structures

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May 9, 2026

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Validity of the Constructed Proofs

The proofs completed in Part One focused on determining whether statements could be verified or disproven through logical reasoning. Some problems were more naturally solved through algebraic manipulation, while others were much easier to approach through counterexamples. In each case, the validity of the proof depended on whether the conclusion followed correctly from the assumptions, definitions, and logical structure being used throughout the argument.

Question 1 focused on proving that an integer N is odd if and only if N^3 is odd. This proof required both directions of the statement to be shown separately. The first direction used a direct proof by expressing an odd integer as $2k+1$. After expanding the cubic expression, the result still took the form of an odd integer, showing that cubing an odd number produces an odd result. For the converse direction, I used a proof by contrapositive rather than proving the statement directly. Showing that an even integer always produces an even cube turned out to be much easier algebraically, while still proving the same logical relationship.

The set-theoretic proofs in Questions 3 and 4 relied heavily on the definitions of subsets, intersections, set differences, and Cartesian products. Those problems mostly involved carefully tracking whether elements belonged to certain sets and then verifying that the relationships remained consistent on both sides of the equation. Question 4 required close attention to subset relationships and element membership to avoid notation mistakes. Other parts of the assignment were more effectively addressed with counterexamples than with direct proofs. In Question 2,

testing a few small integer values quickly showed that some of the converses and contrapositives did not hold universally. That section reinforced how useful counterexamples can be when evaluating universal statements.

Proof Methods Used

Several proof methods were used throughout the assignment, including direct proof, proof by contrapositive, and proof by counterexample. Direct proofs were generally the most straightforward because they started with an assumption and proceeded step by step toward the conclusion using algebraic simplification or formal definitions. This approach worked especially well in Question 1, where it proved that the cube of an odd integer remains odd. Rewriting the integer in the form $2k+1$ naturally guided the proof toward the conclusion after expanding the cubic expression. Direct proofs were also useful in Question 4 when working through the set-theory identities involving intersections, set difference, and Cartesian products.

Question 1 also demonstrated the usefulness of proof by contrapositive. Instead of attempting to prove the converse directly, the statement was rewritten in terms of even integers. Showing that an even integer always produces an even cube made the algebra much easier to organize and follow. Working through this problem showed how choosing a different proof strategy can greatly reduce a proof's complexity while still establishing the same logical relationship.

Counterexamples were another important proof technique used throughout the activity. In Question 2, small integer values were enough to show that certain converses and contrapositives did not hold universally. For example, testing $a=2$ and $b=1$ immediately showed that one of the

statements failed because the hypothesis was true while the conclusion was false.

Counterexamples were also important in Question 4 when disproving the statement involving subset relationships between sets X and Z . Compared to longer algebraic proofs, counterexamples provided a much faster and clearer way to evaluate whether a universal statement could hold in every possible case.

Strengths and Weaknesses of the Proofs

One of the stronger aspects of these proofs was their organization. Most of the problems followed a clear progression where assumptions, definitions, and conclusions connected logically from one step to the next. In Question 1, using the standard forms $2k$ for even integers and $2k+1$ for odd integers helped keep the algebra organized and consistent throughout the proof. In the set-theory proofs from Questions 3 and 4, carefully tracking element membership made it easier to verify subset relationships and avoid logical gaps while working through intersections and Cartesian products.

Another strength was the variety of proof techniques used across the assignment. Some problems, such as proving that the cube of an odd integer remains odd in Question 1, worked naturally as direct proofs. Other problems became much easier once a contrapositive or counterexample was introduced. Question 2 showed how quickly a universal statement could fail once a single contradiction was identified. Working through different proof styles reinforced the idea that proof construction is not always about using one universal method. In many cases, selecting the right proof strategy is part of solving the problem itself.

One weakness of some proofs was readability, particularly in the longer algebraic sections. In Question 1, the algebra became somewhat difficult to follow once the cubic

expression was expanded into multiple terms. Mathematical notation can quickly become confusing if too many simplification steps occur at once or if formatting is inconsistent. Some of the set-theory proofs also required careful attention to notation because symbols involving subsets, intersections, and set membership can easily be mixed up. Looking back, several proofs could likely have been clearer if the hypotheses and conclusions had been identified more explicitly before the formal proof began. Overall, the assignment reinforced the importance of organization, precision, and proof strategy in constructing mathematical arguments.

References

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AI Acknowledgment

Grammarly was used to assist with proofreading and revision. GPT was used to clarify proof structures. All submitted work was reviewed and edited by the author (Zachery Thompson).